A Chromatic Number Algorithm via Inclusion-Exclusion over Independent Sets

Graph Exact Coloring Heuristic Project

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Abstract

We describe and analyze an algorithm that determines the chromatic number of a graph by evaluating a combinatorial quantity derived from inclusion-exclusion over the collection of all independent sets. The quantity, denoted $c_k(G)$, counts the number of ordered k-tuples of independent sets of G whose union is $V(G)$. We prove that $c_k(G) = 0$ if and only if $k < \chi(G)$, so the chromatic number equals the smallest k for which $c_k(G) \neq 0$. We also describe a companion greedy heuristic that computes an upper bound on $\chi(G)$. Both algorithms are implemented in C++17 using the Boost Graph Library. We present experimental results on cycles C_n, complete graphs K_n, the Petersen graph $GP(5, 2)$ and other generalized Petersen graphs, wheel graphs W_n, complete multipartite graphs $K_{s,\ldots,s}$, and edgeless graphs, confirming that the exact algorithm recovers the known chromatic numbers for all tested families. We also report empirical scaling data showing that the algorithm is practical for graphs up to roughly 13–14 vertices on a single core.

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1 Introduction

Graph coloring is one of the most extensively studied problems in combinatorics and theoretical computer science. Given a graph $G = (V, E)$, a *proper k-coloring* assigns one of k colors to each vertex so that no two adjacent vertices share a color. The minimum such k is the *chromatic number* $\chi(G)$. Determining $\chi(G)$ is NP-hard in general [1], yet exact algorithms are essential for small or sparse graphs arising in scientific computing, scheduling, and register allocation.

This paper documents the algorithm implemented in the `graph-exact-coloring-heuristic` project. The core function `c_k` evaluates the inclusion-exclusion sum

$$c_k(G, k) \;=\; \sum_{S \subseteq V} (-1)^{|S|} \left|\{\, I \in \mathcal{F}(G) : I \cap S = \emptyset \,\}\right|^k,$$

where $\mathcal{F}(G)$ is the family of all independent sets of G (including the empty set). We show that this quantity is zero precisely when G is not k-colorable, and hence one can binary-search (or linearly scan) over k to find $\chi(G)$.

The companion greedy algorithm, implemented in `main2.cpp`, repeatedly extracts a greedy independent set and assigns it the next available color, yielding an upper bound that is exact on bipartite and perfect graphs but may exceed $\chi(G)$ in general.

Organization. Section 2 fixes notation and recalls standard facts about graph coloring. Section 3 derives the formula and proves its correctness. Section 4 describes the greedy heuristic. Section 5 outlines the C++ implementation. Section 6 presents experimental results. Section 7 analyzes time and space complexity. Section 8 concludes.

2 Preliminaries

Throughout, $G = (V, E)$ is a simple undirected graph with $n = |V|$ vertices and $m = |E|$ edges. Vertices are labeled $0, 1, \ldots, n-1$.

Definition 1 (Independent set)**.** *A set $I \subseteq V$ is* independent *if no two vertices in I are adjacent. We write $\mathcal{F}(G)$ for the family of all independent sets of G, including $\emptyset$.*

Definition 2 (Proper coloring and chromatic number)**.** *A* proper k-coloring *of G is a function $f : V \to \{1, \ldots, k\}$ with $f(u) \neq f(v)$ whenever $uv \in E$. The* chromatic number $\chi(G)$ *is the minimum k for which a proper k-coloring exists.*

Definition 3 (Independent set cover)**.** *An* ordered k-cover by independent sets *(hereafter, k-IS-cover) is a k-tuple $(I_1, \ldots, I_k) \in \mathcal{F}(G)^k$ such that $I_1 \cup \cdots \cup I_k = V$.*

Note that in a k-IS-cover the sets need not be disjoint, and they may be empty. A proper k-coloring induces a k-IS-cover (the color classes), but a k-IS-cover may have overlapping sets. We will show these notions are equivalent for the *existence* question.

Definition 4 (Chromatic polynomial)**.** *The* chromatic polynomial $P(G, k)$ *counts the number of proper k-colorings of G. It satisfies $P(G, k) = 0$ iff $k < \chi(G)$.*

Well-known chromatic polynomials for families used in experiments:

$$\begin{aligned} P(K_n, k) &= k(k-1)(k-2)\cdots(k-n+1), \\ P(C_n, k) &= (k-1)^n + (-1)^n(k-1), \\ P(E_n, k) &= k^n, \end{aligned}$$

where K_n is the complete graph, C_n the cycle, and E_n the edgeless graph on n vertices.

3 The Exact Algorithm

3.1 The Combinatorial Formula

For a graph G and a positive integer k, define

$$\alpha(S, \mathcal{F}) \;=\; \big|\{\, I \in \mathcal{F}(G) : I \cap S = \emptyset \,\}\big|$$

to be the number of independent sets of G that avoid S entirely. Equivalently, $\alpha(S, \mathcal{F})$ is the total number of independent sets in the subgraph of G induced by $V \setminus S$. The algorithm evaluates

$$c_k(G, k) \;=\; \sum_{S \subseteq V} (-1)^{|S|}\, \alpha(S, \mathcal{F})^k. \tag{1}$$

3.2 Combinatorial Interpretation

Theorem 1. *$c_k(G, k)$ equals the number of ordered k-IS-covers of G.*

Proof. Let $N = |\mathcal{F}(G)|$. The number of ordered k-tuples $(I_1, \ldots, I_k) \in \mathcal{F}(G)^k$ with no constraint is N^k. For a fixed vertex $v \in V$ let A_v be the set of k-tuples in which v is covered by no I_j (i.e. $v \notin I_j$ for all j). Then

$$|A_v| = |\{I \in \mathcal{F}(G) : v \notin I\}|^k = \alpha(\{v\}, \mathcal{F})^k.$$

More generally, for any $S \subseteq V$,

$$\Big|\bigcap_{v \in S} A_v\Big| = |\{I \in \mathcal{F}(G) : S \cap I = \emptyset\}|^k = \alpha(S, \mathcal{F})^k.$$

By inclusion-exclusion, the number of k-tuples in which every vertex is covered by at least one I_j is

$$\sum_{S \subseteq V} (-1)^{|S|}\, \alpha(S, \mathcal{F})^k \;=\; c_k(G, k). \qquad \square$$

3.3 Equivalence with k-Colorability

Theorem 2. *$c_k(G,k) = 0$ if and only if G is not properly k-colorable, i.e. $k < \chi(G)$.*

Proof. ($\Leftarrow$) Suppose $k \geq \chi(G)$. Fix a proper k-coloring $f : V \to \{1, \ldots, k\}$ and let $I_j = f^{-1}(j)$ for each j. Each I_j is an independent set and $\bigcup_j I_j = V$, so $(I_1, \ldots, I_k)$ is a k-IS-cover. Hence $c_k(G,k) \geq 1 > 0$.

($\Rightarrow$) Suppose $k < \chi(G)$. We show no k-IS-cover exists. If $(I_1, \ldots, I_k)$ were a k-IS-cover, define $f(v) = \min\{j : v \in I_j\}$ for each $v \in V$. Since I_j is independent, no two adjacent vertices can share the same j; thus f is a proper k-coloring, contradicting $k < \chi(G)$. Therefore $c_k(G,k) = 0$. □

Corollary 3. $\chi(G) = \min\{\, k \geq 1 : c_k(G,k) \neq 0 \,\}$.

Remark 1. *The quantity $c_k(G,k)$ is not the chromatic polynomial $P(G,k)$ in general. For $k = \chi(G)$, the two coincide for complete graphs (where every k-IS-cover induces a proper coloring immediately), but they differ for most other graphs at values $k > \chi(G)$. For example, $c_k(C_5, 3) = 60$ while $P(C_5, 3) = 30$. Nevertheless, both are zero for $k < \chi(G)$ and nonzero for $k \geq \chi(G)$, which is the only property needed to determine the chromatic number.*

3.4 Algorithm

The full procedure to compute $\chi(G)$ is given in Algorithm 2. It first enumerates all independent sets of G (Algorithm 1) using a power-set sweep with bitset membership testing, then evaluates formula (1) for increasing k.

Algorithm 1 Enumerate all independent sets of G

Require: Graph $G = (V, E)$ with $|V| = n$
Ensure: Set $\mathcal{F}$ of all independent sets (as n-bit integers)

```
F ← ∅
for i ← 0 to 2^n − 1 do
    B ← n-bit representation of i
    if ∀ u ≠ v: B[u] = B[v] = 1 ⇒ uv ∉ E then
        F ← F ∪ {B}
    end if
end forreturn F
```

Bitset Representation

Each independent set is stored as an n-bit integer (a `boost::dynamic_bitset`). The independence check for a candidate set B iterates over all pairs (i, j) with $B[i] = B[j] = 1$ and verifies $ij \notin E$. The subset-avoidance test $I \cap S = \emptyset$ reduces to the bitwise operation $(I \,\&\, S) = 0$.

4 The Greedy Coloring Heuristic

4.1 Independent Set Extraction

The greedy procedure (Algorithm 3) builds an independent set by iterating through the active vertices in index order. For each vertex v selected, it removes v and all of v's current neighbors from the active set.

Algorithm 2 Compute $\chi(G)$ via inclusion-exclusion

Require: Graph $G = (V, E)$ with $|V| = n$
Ensure: Chromatic number $\chi(G)$

```
F ← EnumerateIndepSets(G)
for k ← 1 to n do
    cnt ← 0
    for each S ⊆ V (represented as a bitset) do
        a ← |{I ∈ F : I & S = 0}|                    ▷ count IS avoiding S
        cnt ← cnt + (−1)^|S| · a^k
    end for
    if cnt ≠ 0 then return k
    end if
end for
```

Algorithm 3 Greedy independent set extraction

Require: Graph $G = (V, E)$; active set $U \subseteq V$
Ensure: An independent set $I \subseteq U$

```
I ← ∅
while G has edges do
    v ← first vertex in U
    I ← I ∪ {v}
    Remove v and all neighbors of v from U and from G
end whilereturn I
```

4.2 Greedy Coloring

Algorithm 4 repeatedly calls the IS extractor to peel off color classes one at a time.

Algorithm 4 Greedy graph coloring

Require: Graph $G = (V, E)$
Ensure: Coloring array $col[0 \ldots n-1]$

```
col ← array of size n; k ← 0
while G has edges do
    I ← GreedyIndepSet(G)
    for each v ∈ I do
        col[v] ← k
        Remove all edges incident to v from G
    end for
    k ← k + 1
end whilereturn col
```

4.3 Quality of the Heuristic

The greedy algorithm is well-known to produce at most $\Delta(G) + 1$ colors, where $\Delta(G)$ is the maximum degree [2]. It is exact on bipartite graphs and on chordal graphs (with an appropriate vertex ordering), but can be arbitrarily far from optimal on general graphs. Its primary advantage is $O(n + m)$ running time per color class, making it practical for large sparse graphs.

5 Implementation

The project is written in C++17 and uses the Boost Graph Library (BGL) version 1.60 or later. The graph type is

```
typedef boost::adjacency_list<
    boost::vecS, boost::vecS, boost::undirectedS,
    property<boost::vertex_color_t, int>,
    property<boost::edge_weight_t, int,
             property<boost::edge_name_t, string>>> Graph;
```

Independent sets are stored as `boost::dynamic_bitset<>` objects, which support $O(n/w)$ bitwise AND, OR, and population-count operations where w is the machine word size.

The key functions are:

- `power_set(n, f)`: iterates over all 2^n subsets, calling `f` with each subset represented as a `dynbit`.
- `is_indep_set(G, B)`: checks all pairs of set bits in B for adjacency.
- `gen_ind_set_for_g(G)`: returns $\mathcal{F}(G)$ as a `std::set<dynbit>`.
- `alpha(S, F)`: counts $|\{I \in F : I \,\&\, S = 0\}|$ by a linear scan.
- `c_k(F, G, k)`: evaluates formula (1) using floating-point arithmetic and casts the result to `int`.

Generators for Cycle (C_n) and Generalized Peterson ($GP(n,k)$) graphs extend the abstract base class `GeneratorInterface` and override `generate()`.

Generalized Peterson graph. $GP(n,k)$ has $2n$ vertices. The outer n-cycle connects vertex i to $(i+1) \bmod n$. Spoke edges connect outer vertex i to inner vertex $i+n$. Inner edges connect $i+n$ to $(i+n+k) \bmod n+n$ with a boundary correction.

Build system. The project uses CMake 3.6 or later. The `exact_coloring` target compiles `main.cpp` (exact algorithm demo) and the `tests` target compiles `tests.cpp` (the test suite described in Section 6).

6 Experimental Results

All experiments were run on the `tests` binary compiled with `-std=c++17 --openmp -O2` on Ubuntu 22.04. Table 1 summarizes the chromatic numbers recovered by the exact algorithm.

All 65 test assertions pass, confirming both the zero/nonzero behavior of c_k and, for complete graphs, the exact value of the formula at $k = \chi(G)$.

6.1 Exact Values of c_k at the Chromatic Number

For complete graphs, the formula (1) coincides with the standard chromatic polynomial at $k = \chi(G)$:

The coincidence for K_n can be verified algebraically: since the only independent sets of K_n are $\emptyset$ and the n singletons, $\alpha(S, \mathcal{F}) = n - |S| + 1$ and the sum (1) at $k = n$ evaluates to

$$\sum_{j=0}^{n} \binom{n}{j} (-1)^{n-j} (j+1)^n = n!,$$

which equals $P(K_n, n) = n!$. For cycle C_4 the coincidence is accidental; for C_5 the values differ.

6.2 Parity Property of Odd Cycles

For any odd cycle C_n (n odd) the algorithm correctly returns $\chi = 3$. No odd cycle is 2-colorable because any 2-coloring of a path forces the two endpoints of a cycle to receive the same color, contradicting the closing edge. For $n = 3, 5, 7$ the algorithm confirmed $c_k(C_n, 2) = 0$ and $c_k(C_n, 3) > 0$.

6.3 Petersen Graph

The Petersen graph $GP(5, 2)$ is the canonical example of a 3-chromatic 3-regular graph. It is not planar and has no Hamiltonian cycle. Our algorithm confirmed $\chi(GP(5, 2)) = 3$, with $c_k(GP(5, 2), 2) = 0$ and $c_k(GP(5, 2), 3) > 0$.

6.4 Wheel Graphs

The wheel graph W_n consists of a single hub vertex connected to every vertex of a cycle C_{n-1}. It has n vertices and $2(n-1)$ edges. Because the hub is adjacent to every rim vertex, the hub's color must differ from all colors used on the rim.

- If n is odd, the rim C_{n-1} has an even number of vertices and is therefore bipartite, so $\chi(C_{n-1}) = 2$. The hub requires a third color, giving $\chi(W_n) = 3$.
- If n is even, the rim C_{n-1} is an odd cycle, so $\chi(C_{n-1}) = 3$. The hub must avoid all three rim colors, giving $\chi(W_n) = 4$.

This makes wheel graphs the simplest natural family with $\chi = 4$. We verified $\chi(W_n)$ for $n \in \{5, 6, 7, 8, 9\}$; all 5 tests pass.

6.5 Complete Multipartite Graphs

The complete k-partite graph $K_{s,\ldots,s}$ (with k parts of s vertices each) has $\chi = k$: color each part with its own color (lower bound from the clique-like independence structure; k colors suffice since each part is an independent set). We tested:

- $K_{2,2,2}$ (octahedron): 6 vertices, 12 edges, $\chi = 3$.
- $K_{3,3,3}$: 9 vertices, 27 edges, $\chi = 3$.
- $K_{2,2,2,2}$: 8 vertices, 24 edges, $\chi = 4$.

These graphs have few independent sets relative to 2^n (only subsets within a single part), so the algorithm is fast despite the large edge count.

6.6 Generalized Petersen Family

Beyond $GP(5,2)$ we tested five more members:

- $GP(4,1) \cong Q_3$ (3-cube): bipartite, $\chi = 2$.
- $GP(5,1)$ (pentagonal prism): odd outer cycle, $\chi = 3$.
- $GP(6,1)$ (hexagonal prism): even outer cycle, bipartite, $\chi = 2$.
- $GP(6,2)$: inner triangles force $\chi \geq 3$; algorithm confirms $\chi = 3$.
- $GP(7,2)$: odd outer and inner cycles, $\chi = 3$.

6.7 Scaling Analysis

Table 3 reports wall-clock times for selected larger graphs. The dominant cost is evaluating c_k (formula (1)), which is $O(2^n \cdot |\mathcal{F}|)$ per candidate k. For sparse graphs such as cycles, $|\mathcal{F}|$ grows like the Lucas sequence $L(n) \approx \varphi^n$, so the total cost is roughly $O((2\varphi)^n) \approx O(3.24^n)$, yielding approximately a $10\times$ slowdown per 2 additional vertices. Dense graphs such as $K_{m,m}$ have much smaller $|\mathcal{F}|$ and therefore scale better.

Overflow limitation. The implementation accumulates the inclusion-exclusion sum in a 32-bit `int`. For dense complete graphs K_n with $n \geq 9$, individual terms of the form $(-1)^{|S|}\,\alpha(S,\mathcal{F})^k$ exceed 2^{31}, causing silent overflow. Cycles and bipartite graphs are not affected because their independent-set counts are much smaller. Replacing `int` with `int64_t` would extend the safe range to K_n with $n \lesssim 18$.

7 Complexity Analysis

7.1 Independent Set Enumeration

`gen_ind_set_for_g` calls `power_set`, which iterates over all 2^n subsets. For each subset B, `is_indep_set` checks all pairs of set bits: at most $O(n^2)$ edge queries. Hence the enumeration phase runs in $O(2^n \cdot n^2)$ time and produces a set $\mathcal{F}$ of at most $3^n/\mathrm{poly}(n)$ independent sets (by the Moon–Moser theorem on the maximum number of maximal independent sets in a graph with n vertices [3]).

7.2 Evaluation of c_k

For each candidate k, evaluating formula (1) requires iterating over all 2^n subsets S and, for each S, scanning all $|\mathcal{F}|$ independent sets to compute $\alpha(S,\mathcal{F})$. This costs $O(2^n \cdot |\mathcal{F}| \cdot n/w)$ per value of k, where n/w is the cost of a bitwise AND. Summing over at most n values of k, the total evaluation cost is $O(n \cdot 2^n \cdot |\mathcal{F}| \cdot n/w)$.

7.3 Overall Complexity

The dominating term is the evaluation of c_k, giving

$$T(n) = O\big(n^2 \cdot 2^n \cdot |\mathcal{F}(G)|\big)\,.$$

In the worst case (e.g., the empty graph E_n has $|\mathcal{F}| = 2^n$), this is $O(n^2 \cdot 4^n)$. Empirically (Table 3), sparse graphs such as cycles reach the practical limit around $n = 13$–14 (seconds to minutes), while

dense graphs such as $K_{m,m}$ scale better because $|\mathcal{F}|$ is much smaller than 2^n. The algorithm is intended for exact coloring of small subgraphs; for large graphs the greedy heuristic in `main2.cpp` is the appropriate choice.

7.4 Space Complexity

Storing $\mathcal{F}$ requires $O(|\mathcal{F}| \cdot n/w)$ words, which is $O(3^n \cdot n/w)$ in the worst case.

8 Conclusion

We have described and formally justified the inclusion-exclusion algorithm for computing the chromatic number of a graph implemented in the `graph-exact-coloring-heuristic` project. The key result is that the quantity $c_k(G, k)$ — the number of ordered k-tuples of independent sets covering V — is zero if and only if G is not k-colorable (Theorem 2). Scanning $k = 1, 2, \ldots$ and stopping at the first nonzero value thus yields $\chi(G)$ exactly.

The algorithm is exponential in $|V|$ and is therefore intended for small graphs or for exact coloring of subgraphs within a larger heuristic pipeline. The companion greedy coloring algorithm runs in linear time and provides a fast upper bound useful as an initialization strategy or a standalone heuristic for large instances.

Future directions include: (i) exploiting symmetry to prune the power-set enumeration, (ii) integrating branch-and-bound to improve worst-case behavior, and (iii) leveraging OpenMP (already supported in the build) to parallelize the inner α-counting loop.

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Table 1: Chromatic numbers computed by the exact algorithm (65 assertions, all passing).

Graph	Description	$\lvert V\rvert$	$\lvert E\rvert$	χ (computed)	Known χ
Cycle graphs					
C_3	Triangle / K_3	3	3	3	3
C_4	4-cycle (square)	4	4	2	2
C_5	5-cycle (pentagon)	5	5	3	3
C_6	6-cycle (hexagon)	6	6	2	2
C_7	7-cycle	7	7	3	3
Complete and edgeless graphs					
K_1	Single vertex	1	0	1	1
K_2	Single edge	2	1	2	2
K_3	Triangle	3	3	3	3
K_4	Complete graph	4	6	4	4
E_1	Isolated vertex	1	0	1	1
E_3	Three isolated vertices	3	0	1	1
Bipartite graphs					
$K_{2,3}$	Complete bipartite	5	6	2	2
Wheel graphs $W_n = \text{hub} + C_{n-1}$; $\chi = 3$ if n odd, $\chi = 4$ if n even					
W_5	Hub $+ C_4$ (even rim)	5	8	3	3
W_6	Hub $+ C_5$ (odd rim)	6	10	4	4
W_7	Hub $+ C_6$ (even rim)	7	12	3	3
W_8	Hub $+ C_7$ (odd rim)	8	14	4	4
W_9	Hub $+ C_8$ (even rim)	9	16	3	3
Complete multipartite graphs; $\chi(K^{(k)}_{s,\ldots,s}) = k$					
$K_{2,2,2}$	Octahedron	6	12	3	3
$K_{3,3,3}$	Complete 3-partite	9	27	3	3
$K_{2,2,2,2}$	Complete 4-partite	8	24	4	4
Generalized Petersen graphs $GP(n,k)$					
$GP(4,1)$	3-cube Q_3 (bipartite)	8	12	2	2
$GP(5,1)$	Pentagonal prism	10	15	3	3
$GP(5,2)$	Petersen graph	10	15	3	3
$GP(6,1)$	Hexagonal prism (bipartite)	12	18	2	2
$GP(6,2)$	—	12	18	3	3
$GP(7,2)$	—	14	21	3	3

Table 2: Values of $c_k(G, \chi(G))$ vs. the chromatic polynomial $P(G, \chi(G))$.

Graph	χ	$c_k(G, \chi)$	$P(G, \chi)$	Match?
K_1	1	1	1	Yes
K_2	2	2	2	Yes
K_3	3	6	6	Yes
K_4	4	24	24	Yes
C_3	3	6	6	Yes
C_4	2	2	2	Yes
C_5	3	60	30	No

Table 3: Wall-clock time for larger graphs (Release build, single core, Ubuntu 22.04).

Graph	Family	$\lvert V\rvert$	$\lvert E\rvert$	χ	Time
C_8	Cycle	8	8	2	1 ms
C_9	Cycle	9	9	3	7 ms
C_{10}	Cycle	10	10	2	24 ms
C_{11}	Cycle	11	11	3	86 ms
C_{12}	Cycle	12	12	2	168 ms
C_{13}	Cycle	13	13	3	786 ms
C_{14}	Cycle	14	14	2	$\approx$3 s
C_{16}	Cycle	16	16	2	$\approx$33 s
K_5	Complete	5	10	5	$<$1 ms
K_6	Complete	6	15	6	$<$1 ms
K_7	Complete	7	21	7	$<$1 ms
K_8	Complete	8	28	8	1 ms
$K_{4,4}$	Bipartite	8	16	2	1 ms
$K_{5,5}$	Bipartite	10	25	2	9 ms
$K_{6,6}$	Bipartite	12	36	2	69 ms
$K_{7,7}$	Bipartite	14	49	2	536 ms
$GP(6, 2)$	Gen. Petersen	12	18	3	135 ms
$GP(7, 2)$	Gen. Petersen	14	21	3	1333 ms
$GP(8, 3)$	Gen. Petersen	16	24	2	$\approx$15 s